DATA STRUCTURES LABORATORY			
Course Code	**1BCSL306**	Scheme	2025
Type of Course	PCCL	Semester	3
Teaching Hours/Week (L:T:P)	0:0:2	CIE Marks	50
Total Hours of Pedagogy L:T:P:SL:TW	0:0:28:2:30	SEE Marks	50
Credits	1	Total Marks	100
Examination type (SEE)	Practical	Exam Hours	03

Course outcome (Course Skill Set)

At the end of the course, the student will be able to:

1. Develop programs to implement fundamental and linear data structures and their operations for solving computational problems.
2. Develop programs using non-linear data structures and hashing techniques to solve computational problems.
3. Select appropriate data structure to organize, process, and retrieve data while solving real-world problems.
4. Build experimental documents involving data structures, algorithms, program design, implementation and results with inferences.

Note:

1. The laboratory syllabus consists of PART-A and PART-B. While PART-A has 6 conventional experiments, PART-B has 6 typical open-ended experiments. The maximum marks for the laboratory course are 100.
2. Both PART-A and PART-B are considered for CIE and SEE.
3. Students have answer 1(one) question from PART-A and 1(one) question from PART-B.
   a. The questions set for SEE shall be from among the experiments under PART-A. It is evaluated for **70 marks** out of the maximum 100 marks.
   b. The open-ended question set for SEE shall be any other open-ended question and not selected from the experiments under PART-A. It shall be evaluated for **30 marks.**
4. For Continuous Internal Evaluation (CIE) during the semester, classwork shall typically include open-ended questions from **Part-B**, along with other similar questions designed to enhance the students' skills.

PART – A
FIXED SET OF EXPERIMENTS

1. Develop a Program in C for the following:

   a. Define a record (structure) titled Book with the following fields: Book_ID, Title, Author, Price, Availability Status (Available/Issued).

   b. Dynamically allocate memory to store the details of N books, where N is entered by the user.

   c. Implement the following functions:

      i. create() – To allocate memory and read the details of N books and store them in the allocated memory.

      ii. display() – To display the details of all books available in the library.

      iii. search() – To search for a book using Book ID and display the book details if found.

      iv. issueBook() – To issue a book by updating its availability status and display an appropriate message.

      v. returnBook() – To return an issued book by updating its availability status and display an appropriate message.

   d. Build a menu-driven interface with the following options:

      i. Add Book Records.

ii. Display all Book Records.
iii. Search Book by Book ID.
iv. Issue a Book.
v. Return a Book.
vi. Exit.

2. Develop a C program for processing sparse matrices using an appropriate data structure. The program shall accept suitable sparse matrix data, use an appropriate representation, and perform operations such as addition and transpose. Demonstrate the correctness of the solution using suitable test cases.

3. Develop a menu driven Program in C for the following operations on STACK of Integers (Array Implementation of Stack with maximum size MAX >15).
   a. Push an Element on to Stack.
   b. Pop an Element from Stack.
   c. Demonstrate Overflow and Underflow situations on Stack.
   d. Display the status of Stack at any given point of time.
   e. Exit.

   Support structured program with appropriate functions for each of the above operations.

4. Develop a program in C for simulating a Printer Queue using Queue data structure with the following operations:
   a. Add a print job to the Printer Queue.
   b. Process a print job from the Printer Queue (delete).
   c. Display the number of print jobs currently waiting in the Queue.
   d. Demonstrate Queue overflow and underflow conditions.
   e. Exit.

5. Develop a Program in C for the following operations on Singly Circular Linked List (SCLL) with header nodes.
   a. Read the polynomial: degree of the polynomial, terms (coefficient and power).
   b. Represent the Polynomial in one variable based on degree.
   c. Find the sum of two polynomials POLY1 and POLY2 and store the result in POLYSUM.

   Support the program with appropriate functions for each of the above operations.

6. Develop a Program in C for the following:
   a. Create a Binary Tree (createTree) by inserting integer elements as keys entered by the user in level order.
   b. Implement the following functions:
      i. createTree() – Create a binary tree through key Insertions.
      ii. preorder() – Display the elements using Preorder Traversal.

iii. inorder() – Display the elements using Inorder Traversal.

iv. postorder() – Display the elements using Postorder Traversal.

c. Develop a menu-driven interface with the following options:

i. Create Binary Tree.

ii. Display Preorder Traversal.

iii. Display Inorder Traversal.

iv. Display Postorder Traversal.

v. Exit.

PART – B

OPEN ENDED EXPERIMENTS

Open-ended experiments are a type of laboratory activity where the outcome is not predetermined and students are given the freedom to explore, design, and conduct the experiment based on the problem statements as per the concepts defined by the course coordinator. It encourages creativity, critical thinking, and inquiry-based learning.

1. A compiler design team needs a tool to convert arithmetic expressions from infix notation to postfix notation for efficient code generation. The expression contains alphanumeric operands, operators (+, -, *, /, %, ^), and brackets: (,), {, }, [,]. Develop a C program that supports the following::

   a. Read an infix expression.

   b. Validate the expression for correct bracket nesting and syntax.

   c. Convert the valid infix expression to postfix notation.

   Note: An input expression is valid if:

   • Open brackets must be closed by the same type of brackets.

   • Open brackets must be closed in the correct order.

   • Every close bracket has a corresponding open bracket of the same type.

   • Operators must have valid operands on both sides.

2. A University wants to manage student records effectively. Each student record contains USN, Name, Semester (1-8), and Department. Choose suitable data structure (Singly Linked List/ Doubly Linked List/Circular Linked Lists) and develop a menu-driven C program to manage these records with the following operations:

   a. Create a list of student records (choose any insertion method: beginning /end).

   b. View all student records.

   c. Display student records for a given semester.

   d. Insert the new student in the list after the given student's name.

   e. Delete a student record (based on USN or Name)

   Note: For CIE Test and SEE, any ONE option from (c) to (e) may be asked along with options (a) & (b).

3. A manufacturing unit needs a system (repository) to manage machine parts used in assembly. Each part has:

   • Part Number (alphanumeric, e.g., "P1001", "BOLT23")

   • Part Name (e.g., "Bolt", "Wheel")

- Model Number (e.g., "M2024", "X100")
- Description (e.g., "Stainless steel ")

Develop a C program to implement the following three operations.

a. Add a new part to the inventory/repository

b. Remove an obsolete part (by part number or model number)

c. Search and display part details (by part number or model number)

All three operations happen frequently without any order. Choose suitable data structure that results in the best performance.

4. Develop a C program by selecting suitable data structure to manage hierarchical data/information having keys (numeric). The program shall provide appropriate operations such as insertion of information, traversal (display of all information), quick search for information based on the key. Demonstrate the solution using relevant test cases.

5. Develop a C program to store employee records having fields: emp-no, name, designation, department) using suitable Indexing mechanism. The program shall support record management operations such adding new employee, search for employee details based on emp-no, remove employee record based on emp-no, and display employee records based on department. Demonstrate the correctness of the implementation using appropriate test cases.

6. A transportation department wants to analyse connectivity between cities. Each city has a City Name and cities are interconnected (not necessarily all) with roads. Develop a menu-driven C program with the following operations:

a. Create a graph representing N cities with interconnecting roads.

b. Display the graph representation.

c. Print all cities reachable by road from a given city..

d. Check whether any two given cities are connected via road.

Suggested Learning Resources: (Text Book/ Reference Book/ Manuals):

Text books:

1. Ellis Horowitz, Sartaj Sahni and Susan Anderson-Freed, Fundamentals of Data Structures in C, 2nd Ed, Universities Press, 2023
2. Reema Thareja, Data Structures using C, 3rd Ed, Oxford press, 2023.

Reference books / Manuals:

1. A M Tenenbaum, Data Structures using C, PHI, 1989.
2. Seymour Lipschutz, Data Structures Schaum's Outlines, McGraw Hill, 2014.

Web links and Video Lectures (e-Resources):

1. GNU-GCC Documentation: https://gcc.gnu.org/
2. Virtual Labs on Data Structures-1: https://ds1-iiith.vlabs.ac.in/List%20of%20experiments.html
3. DSA Tutorial: https://www.geeksforgeeks.org/dsa/dsa-tutorial-learn-data-structures-and-algorithms/
4. NPTEL Course on DSA: https://nptel.ac.in/courses/106106130
5. NPTEL Course on Introduction to C Programming: https://nptel.ac.in/courses/106104128

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. Engineering tool usage for the conduction of experiment
2. Demonstration through ICT tools
3. Use of virtual labs

Assessment Structure:

The assessment for each course is equally divided between Continuous Internal Evaluation (CIE) and the Semester End Examination (SEE), with each component carrying **50% weightage** (i.e., 50 marks each).

The CIE marks awarded shall be based on the continuous evaluation of the laboratory report using a defined set of rubrics. Each experiment report can be evaluated for 30 marks. Total marks for the all report shall be scaled down to 30 marks. The laboratory test (duration 03 hours) at the end of the last week of the semester /after completion of all the experiments (whichever is early) shall be conducted for 50 marks and Marks scored shall be scaled down to 20 marks. Final average of marks for report and test shall be out of 50 marks.
For both CIE and SEE, the student is required to conduct one experiment each from both Part A and Part B.

Rubrics for CIE – Continuous assessment:

Criteria/Scale	Superior (5)	Good (4)	Fair (3)	Needs Improvement (2)	Marginal (1)
Fundamental Knowledge: Understanding the problem statement and selection of appropriate data structure [CO1, CO2, CO3] [PO1, PO2]	Demonstrates in-depth understanding of data structure concepts and correctly identifies the most suitable data structure for the given problem.	Demonstrates good understanding of data structure concepts and selects an appropriate data structure with minor guidance.	Demonstrates satisfactory understanding and selects a suitable data structure for the problem.	Demonstrates partial understanding and selects an inappropriate or partially suitable data structure.	Unable to understand the problem completely or identify an appropriate data structure.
Design of Algorithm and Data Structure [CO1, CO2, CO3] [PO2, PO3]	Develops multiple algorithmic solutions, compares them, and justifies the best design with appropriate data	Develops more than one design but provides limited justification for selecting the best solution.	Develops a correct algorithm and data structure with basic explanation.	Develops an incomplete algorithm or partially correct design.	Unable to design a suitable algorithm or perform data structure operations suitably.

	structure selection.				
Implementation (Program Coding) [CO1, CO2, CO3] [PO5, PO5]	Implements the selected data structure efficiently using appropriate C constructs with correct operations and modular programming practices.	Implements the selected data structure correctly and explains the implementation clearly.	Implements the required data structure with minor coding issues.	Implements the program with significant errors or incomplete operations.	Unable to implement the required data structure program partially.
Program Debugging and Testing [CO1, CO2, CO3] [PO4, PO5]	Successfully compiles, debugs, and validates the program for all test cases without syntax, semantic, or logical errors.	Identifies and rectifies errors independently with good understanding of debugging.	Corrects most errors with partial understanding of debugging.	Corrects errors only with assistance and limited understanding.	Unable to debug or execute the program successfully.
Results and Analysis [CO1, CO2, CO3] [PO4]	Executes the program using various test cases, validates all data structure operations, and provides detailed analysis with appropriate conclusions.	Executes all test cases and explains the results correctly.	Executes most test cases and provides basic analysis.	Executes the program but provides limited or incorrect analysis.	Unable to verify or interpret the program output correctly.
Demonstration and Documentation [CO4] [PO8, PO9]	Demonstration is excellent. Lab record is well-organized with problem statement, algorithm, data structure representation, program, output, observations, and conclusions presented using proper formatting.	Demonstration is good. Lab record is organized with all major sections and appropriate formatting.	Demonstration is satisfactory. Lab record contains most required sections with acceptable formatting.	Demonstration is incomplete. Lab record has missing sections or inadequate formatting.	The demonstration is poor. Lab record is incomplete, improperly formatted, or not submitted on time.

Rubrics for SEE / CIE Test:

Criteria /Scale	Superior (5)	Good (4)	Fair (3)	Needs Improvement (2)	Marginal (1)
Fundamental Knowledge (2) [CO1, CO2] [PO1]	Demonstrates excellent knowledge of data structures, their operations, and applications.	Demonstrates good knowledge of major data structure concepts.	Demonstrates average understanding of basic concepts.	Demonstrates limited conceptual understanding.	Does not understand the fundamental concepts.
Understanding of Problem Definition (1) [CO1, CO2, CO3] [PO2]	Clearly understands the problem and identifies the most appropriate data structure.	Understands the problem with minor clarification.	Demonstrates partial understanding of the problem.	Demonstrates minimal understanding of the problem.	Unable to understand the problem statement.
Design and Implementation (3) [CO1, CO2, CO3] [PO3]	Designs an efficient algorithm and correctly implements the selected data structure using appropriate C constructs.	Designs and implements the required solution with minor issues.	Implements the core algorithm with partial correctness.	Partially implements the solution with significant errors.	Unable to design or implement the solution.
Results and Analysis (2) [CO1, CO2, CO3] [PO4]	Successfully executes the program using multiple test cases, validates all operations, and provides accurate analysis.	Executes the program correctly and provides good analysis.	Executes the program with limited test cases and basic analysis.	Executes the program but cannot properly analyze the results.	Unable to execute or analyze the program.
Communication (Viva Voce) (2) [CO1, CO2, CO3, CO4] [PO8, PO9]	Clearly explains the working of the selected data structure, algorithm, implementation, and application using correct	Explains the concepts with good communication and correct terminology.	Explains basic concepts with average communication skills.	Provides incomplete explanations with incorrect terminology.	Unable to explain the implementation or answer questions satisfactorily.

	technical terminology.				

- To qualify and become eligible to appear for SEE, in the **CIE component**, a student must secure **a minimum of 40% of 50 marks**, i.e., **20 marks**.
- To pass the **SEE component**, a student must secure **a minimum of 35% of 50 marks**, i.e., **18 marks**.
- A student is deemed to have **successfully completed the course** if the **combined total of CIE and SEE is at least 40 out of 100 marks**.